

bpp keyboard, mouse and menus

Part 1 of 2: bpp itself · Part 2 covers the keys of each of your simulations · October 9, 2026

Taken from bpp's source and from libQGLViewer 2.8's own help table (the H key). Where bpp and QGLViewer both use a key, the list says which one wins. A key that the running script binds wins over both.

Esc in the 3D view quits bpp straight away (QGLViewer's "exit"), unless an embedded POV-Ray render is running, when it cancels the render. Unsaved script changes still ask first.

Which keys go where

- **Menu shortcuts** (Ctrl+..., F6, F11, F12) work anywhere in the window.
- **3D-view keys** need the 3D view to have focus: click it first.
- **Script keys:** the order is the script's onKey hook, then its addShortcut keys, then bpp's own, then QGLViewer's.
- **Keys another pane doesn't use** are passed on to the 3D view. Typing S in the read-only Debug pane, or F1 in the script editor, acts as if pressed in the 3D view.

Menus

File

Ctrl+N	New
Ctrl+O	Open...
(none)	Reload file: read the open file again from disk
Ctrl+S	Save
Ctrl+A	Save as... (in the script editor Ctrl+A selects all instead)
(none)	Recent files: 1, 2, 3...
Ctrl+Q	Exit

Edit

F12	Preferences... (see the last section)
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View

F11	Full screen on/off
(none)	Show constraints: draw hinge axes and other constraints
(none)	Shadows: cast shadows from the main light

POV-Ray

Each item has a tooltip describing it; the choices are saved in includes/settings.inc.

(toggle)	Enable / disable the LightSys rendering system
(choice)	Radiosity: off, outdoor low, outdoor high
(choice)	LightSys light preset: daylight or cool fluorescent

(choice)	CIE white point: off, illuminant B, daylight film, indoor film
(toggle)	Secondary LightSys fill light
(toggle)	Area light (soft shadows)
(choice)	Background: none (sky), black, white
(choice)	Clouds / sky: off, simple plane, fast realistic, gradient dome
(toggle)	Ground plane
(choice)	Ground plane texture: plain, grass, glossy white
(choice)	Photon mapping: off, normal, fine
(toggle)	Use embedded VFE for Quick Render (only in builds with VFE)
Help	
(none)	Homepage, Chat, Issues, Wiki (open in the browser); About

Toolbars

From left to right. Hover over a button for its tooltip.

File	New, Open, Reload, Save, Save as
Edit	Preferences
Simulation	Start/stop (Ctrl+C), Restart (Ctrl+R), Toggle deactivation (the D key), Toggle POV export (the P key)
View	Reset the camera to its starting view (the C key); the render size list (view size, 720p, 1080p, 4K, 8K, A4 300/600 dpi, portrait...); Quick render (F6)
Window	Full screen (F11)

Quick render (F6) renders the current frame with POV-Ray at the chosen size and saves a PNG on your Desktop, named bpp-date-time-scene-frame.png.

3D view: keys

bpp's own

S	Start/stop the simulation
R	Run the script again from the start
D	Deactivation (sleeping) on/off: off simulates everything all the time, much slower
P	POV-Ray export on/off: every frame is saved for POV-Ray (slow; bpp says when it turns on)
C	Reset the camera to its starting view
Tab	Single view ↔ four views (perspective, top, front, right)
F1 / F2	Previous / next object, in scripts that set v:cycleObject (otherwise nothing)
Space	During an embedded POV-Ray render: pause/resume it (otherwise QGLViewer's, below)
Esc	During an embedded POV-Ray render: cancel it (otherwise quits bpp, see the top)

QGLViewer's (libQGLViewer 2.8)

Space	Camera mode: revolve (orbit) ↔ fly
Return	Start/stop the animation
Arrows	Move the camera left, right, up, down
A	World axis on/off
G	XY grid on/off
F	Frame-rate display on/off
Shift+?	On-screen text (such as the frame rate) on/off
Alt+Return	QGLViewer's own full screen
H	QGLViewer's help window (keys, mouse, camera paths)
Esc	Quit
F3–F10	Play a camera path or go to a saved position
Alt+F3–F10	Add a key frame to that path (or save a position)
Alt + Fx twice	Holding Alt, press the key twice: delete that path or position

Taken by bpp: S (QGLViewer's stereo), C (its camera-path display), Ctrl+C (its copy-snapshot: in bpp, start/pause), Ctrl+S (its screenshot: in bpp, save the file), F1, F2, F6, F11 and F12 (its camera paths). Alt+F4 normally closes the window, not a path.

3D view: mouse

Revolve (orbit) mode, the default

Left drag	Rotate the camera around the pivot point
Right drag	Move (pan) the camera
Middle drag or wheel	Zoom
Left double-click	Align the camera with the nearest axis
Right double-click	Centre the scene
Middle double-click	Show the entire scene
Shift+Right click	Set the pivot point (the point the camera turns around)
Shift+Middle drag	Zoom in on the region you draw
R + Left drag	Rotate in the screen plane (but R also re-runs the script in bpp)
Z + Left click	Zoom in on that pixel
Z + Right click	Zoom to fit the scene

Fly mode (Space)

Left button / wheel	Move forward
Right button	Move backward
Middle drag	Look around
R + Left drag	Roll the camera

Four-view layout (Tab)

Any drag	In the top, front and right views: pan that view
Wheel	In those views: zoom 10% a notch
Perspective view	The usual mouse, as above

QGLViewer's Shift+Left (select) and Ctrl+mouse (move a manipulated frame) do nothing in bpp: it has neither selection nor manipulated frames.

Mouse-over (hover) tooltips

- **How:** rest the pointer, with no button held, on an object in the 3D view for half a second.
- **What:** the script's tooltip appears by the pointer. It follows a moving object and updates ten times a second.
- **Hiding it:** moving the mouse, a button, the wheel or a key hides it. Hovering never takes a click, so the camera controls work as usual.
- **Which demos:** only scripts that set `v:onHover`. Of yours: the Hover example, Rattleback, Dzhaniybekov and Gomboc Variety.
- **Elsewhere:** toolbar buttons, POV-Ray menu items and Parameters rows show tooltips too. A parameter's tooltip (its description) follows the mouse along the row.

Script editor (LUA Script pane)

Every change you make re-runs the script, so the scene follows your edits.

Ctrl+S	Save
Ctrl+1 / 2 / 3	Open / save / save as (the editor's own)
Ctrl+Space	Completion list: words in the script, Lua keywords, and the language server's suggestions when one is set in Preferences
Return	In the completion list: insert the choice
Up/Down, PgUp/PgDn	In the completion list: move, showing each item's documentation
Ctrl+F	Find bar (again: select the search text)
Return / Shift+Return	In the find bar: next / previous match
Esc	Close the find bar
Ctrl+Z / Ctrl+Shift+Z	Undo / redo
Ctrl+X / C / V	Cut / copy / paste
Ctrl+A	Select all

Find matches are marked in amber along the scroll bar.

The other panes

Command Line	Type a line and press Return. It goes to the script's <code>v.onCommand</code> hook, if it has one (the Formations demos run it as Lua, e.g. <code>addObstacleAt(50, 0, -30)</code>). Up/Down step through earlier lines.
Parameters	Sliders and check boxes for the script's settings; a tooltip gives each one's description. When the pane floats, right-click → Dock to Main Window puts it back.
Shortcuts	The running script's own keys (its help text).
Debug	The script's printed output and errors.
Camera Info	The current camera as Lua lines (<code>v.cam.pos</code> , <code>v.cam.look</code> , <code>v.cam.up</code>) to paste into a script.
Any pane	Drag its title bar to move it, or use its buttons to float or close it.
Status bar	Messages, and the frame number at the right.

Other input devices

3D mouse (SpaceNavigator)	Moves the camera when connected. Button 1 resets the view; buttons 2 and 3 switch between fly and object (orbit) modes. The settings are in Preferences → SpaceNavigator.
Joystick / gamepad	Read by scripts through <code>v.onJoystick</code> ; none of your demos use it.
Preferences (F12)	
GUI	Font and size; save the window position on exit; reload the last file on start
Lua	Package path; language server (for completion)
POV-Ray	Executable; export directory; quick-render options
OpenSCAD	Executable
SpaceNavigator	Navigation mode (fly or object), forward/back direction, lock horizon, auto fly speed, show orbit axis, pan camera view

Command line (terminal)

bpp file.lua	Open a script in the window
-f, --file file	Run a script without the window
-l, --lua expr	Run a Lua expression without the window
-i, --stdin	Run Lua read from standard input
-n, --frames n	How many frames to simulate (default 10)
-e, --export	Export the frames for POV-Ray
-V, --verbose	More output
-r, --report-load	Say which script would be loaded, and exit
-h / -v	Help / version